

OLS as a Statistical Estimator

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0. Introduction

In a parametric linear setting, we have:

$$y = X\beta + \varepsilon$$

where y and $\varepsilon \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times k}$, and $\beta \in \mathbb{R}^k$ are the parameters of interest. For $X\beta = \mathbb{E}[y|X]$ to hold, we assume $\mathbb{E}[\varepsilon|X] = 0$, which is the **strict exogeneity** assumption. Specifically, $\{(x_i, \varepsilon_i)\}_i$ are **i.i.d.**

Corollary

If $\mathbb{E}[\varepsilon|X] = 0$, then

$$\mathbb{E}[\varepsilon] = 0.$$

Normally, in this context, we often want to estimate β and $\text{Var}[\varepsilon|X]$.

After deriving the **closed form solution** for $\hat{\beta}^{\text{OLS}} = (X'X)^{-1}X'y$, which is a **statistical estimator** for β , we want to know its statistical properties, e.g., whether it is **unbiased**, **consistent**, what its **conditional variance** is, and whether it is **asymptotically normal**.

1. Unbiasedness and Consistency

When we have an estimator $\hat{\beta}$ for β , we want to know how good it is, and typically there are three kinds of criteria: unbiasedness, consistency, and **efficiency**.

Unbiasedness is to check whether $\mathbb{E}[\hat{\beta}] = \beta$ and consistency aims to see whether $\hat{\beta} \xrightarrow{p} \beta$.

(i) **Unbiasedness**

Theorem (OLS is an unbiased estimator)

$$\mathbb{E}[\hat{\beta}^{\text{OLS}} \mid X] = \beta$$

Proof

$$\begin{aligned}\mathbb{E}[\hat{\beta}^{\text{OLS}} \mid X] &= \mathbb{E}[(X'X)^{-1}X'y \mid X] \\ &= \mathbb{E}[(X'X)^{-1}X'(X\beta + \varepsilon) \mid X] \\ &= \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X] \\ &= \beta\end{aligned}$$

Note that as $\hat{\beta}^{\text{OLS}} = \beta + (X'X)^{-1}X'\varepsilon$, we only need the condition that $\mathbb{E}[(X'X)^{-1}X'\varepsilon] = 0$ to ensure the unbiasedness. $\mathbb{E}[\varepsilon \mid X] = 0$ is a stricter condition and sufficient for that.

Corollary

$$\mathbb{E}[\hat{\beta}^{\text{OLS}}] = \beta$$

(ii) Consistency**Theorem**

$$\hat{\beta}^{\text{OLS}} \xrightarrow{p} \beta$$

Proof

$$\begin{aligned}\hat{\beta}^{\text{OLS}} &= \beta + (X'X)^{-1}X'\varepsilon \\ &= \beta + \left(\frac{X'X}{n}\right)^{-1} \left(\frac{X'\varepsilon}{n}\right).\end{aligned}$$

To derive the convergence in probability of $\hat{\beta}^{\text{OLS}}$, we need to derive the convergence of $\left(\frac{X'X}{n}\right)^{-1}$ and $\frac{X'\varepsilon}{n}$.

We first assume that $\mathbb{E}[x_i x_i'] = Q \in \mathbb{R}^{k \times k}$ exists and is nonsingular. Since $\{(x_i, \varepsilon_i)\}$ are **i.i.d.**, $\{x_i x_i'\}_i$ are i.i.d. as well, so by **Khinchine's WLLN**,

$$\frac{X'X}{n} = \frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q,$$

and since matrix inversion is continuous at a nonsingular Q , by the **continuous mapping theorem**,

$$\left(\frac{X'X}{n} \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Regarding $\frac{X'\varepsilon}{n} = \frac{1}{n} \sum_i x_i \varepsilon_i$, as $\{(x_i, \varepsilon_i)\}$ are [i.i.d.](#) samples, $\{x_i \varepsilon_i \in \mathbb{R}^k\}_i$ are i.i.d. as well. What's more, by the law of iterated expectations,

$$\mathbb{E}[x_i \varepsilon_i] = \mathbb{E}[x_i \mathbb{E}[\varepsilon_i | x_i]] = 0,$$

so by [Khinchine's WLLN](#),

$$\frac{X'\varepsilon}{n} \xrightarrow{p} 0.$$

Finally, both factors converge in probability to constants, so the pair converges jointly in probability and the [continuous mapping theorem](#) applies to their product:

$$\hat{\beta}^{\text{OLS}} = \beta + \left(\frac{X'X}{n} \right)^{-1} \left(\frac{X'\varepsilon}{n} \right) \xrightarrow{p} \beta + Q^{-1} \cdot 0 = \beta.$$

Note that the convergence:

$$\frac{1}{n} \sum_i x_i x_i' \xrightarrow{p} \mathbb{E}[x_i x_i']$$

can be understood as entry-wise.

2. Asymptotic Normality

(i) Motivation

Once we obtain the estimator $\hat{\beta}^{\text{OLS}}$, the most common first test is whether the true parameter β_j where $1 \leq j \leq k$ differs from zero:

$$H_{0,j} : \beta_j = 0 \quad \text{against} \quad H_{1,j} : \beta_j \neq 0.$$

Notice that we are going to test the parameter one by one, i.e., we will have k null hypotheses, and each associates to one scalar parameter β_j , rather than jointly as:

$$H_0 : \beta = 0 \quad \text{against} \quad H_1 : \beta \neq 0.$$

What we want to do is to derive the asymptotic distribution of $\hat{\beta}_j^{\text{OLS}}$. We know that it is consistent: $\hat{\beta}_j^{\text{OLS}} - \beta_j \xrightarrow{p} 0$, and if we use the central limit theorem to derive that:

$$\sqrt{n}(\hat{\beta}_j^{\text{OLS}} - \beta_j) \xrightarrow{d} N(0, v_j),$$

then we can utilize the statistic

$$t_j = \frac{\sqrt{n}\hat{\beta}_j^{\text{OLS}}}{\sqrt{\hat{v}_j}} \xrightarrow{d} N(0, 1) \quad \text{under } H_{0,j} : \beta_j = 0$$

to do the hypothesis testing. Note that $\hat{v}_j$ should be consistent for v_j .

(ii) Asymptotic Distribution

Theorem (Asymptotic Normality)

$$\sqrt{n}(\hat{\beta}^{\text{OLS}} - \beta) \xrightarrow{d} N(0, Q^{-1}\Omega Q^{-1})$$

where $Q = \mathbb{E}[x_i x_i']$ and $\Omega = \mathbb{E}[\varepsilon_i^2 x_i x_i']$.

Proof

As

$$\hat{\beta}^{\text{OLS}} = \beta + (X'X)^{-1}(X'\varepsilon),$$

multiplying through by $X'X$ and rescaling by n gives

$$\sqrt{n}\left(\frac{X'X}{n}\right)(\hat{\beta}^{\text{OLS}} - \beta) = \sqrt{n}\left(\frac{X'\varepsilon}{n}\right).$$

We already have $\mathbb{E}[x_i \varepsilon_i] = 0$, so $\text{Var}[x_i \varepsilon_i] = \mathbb{E}[\varepsilon_i^2 x_i x_i'] = \Omega$, which we assume exists and is finite. As $\{(x_i, \varepsilon_i)\}$ are i.i.d., by Lindeberg–Lévy's CLT,

$$\sqrt{n}\left(\frac{X'\varepsilon}{n} - 0\right) \xrightarrow{d} N(0, \mathbb{E}[\varepsilon_i^2 x_i x_i']) = N(0, \Omega).$$

Therefore,

$$\sqrt{n}\left(\frac{X'X}{n}\right)(\hat{\beta}^{\text{OLS}} - \beta) \xrightarrow{d} N(0, \Omega).$$

Furthermore, by Khinchine's WLLN and the continuous mapping theorem,

$$\left(\frac{X'X}{n}\right)^{-1} \xrightarrow{p} \mathbb{E}[x_i x_i']^{-1} = Q^{-1},$$

and this limit is a constant matrix, so by Slutsky's theorem,

$$\sqrt{n}(\hat{\beta}^{\text{OLS}} - \beta) \xrightarrow{d} N(0, Q^{-1}\Omega Q^{-1}),$$

where the two outer factors are both Q^{-1} because Q is symmetric.

3. The Variance of the OLS Estimator and the Test Statistic

(i) The Statistic to Compute

Write $V = Q^{-1}\Omega Q^{-1}$ for the asymptotic variance matrix. The scalar v_j is its j -th diagonal entry, $v_j = V_{jj}$, so

$$t_j = \frac{\sqrt{n}\hat{\beta}_j^{\text{OLS}}}{\sqrt{[\hat{Q}^{-1}\hat{\Omega}\hat{Q}^{-1}]_{jj}}}.$$

The estimator

$$\hat{Q} = \left(\frac{1}{n} \sum_i x_i x_i' \right) = \frac{1}{n} X' X$$

is consistent, and the main difficulty is to find the consistent estimator $\hat{\Omega}$.

(ii) Conditional Homoscedasticity

Definition (Homoscedasticity)

The error term is **homoscedastic** when its conditional variance does not vary with the regressors:

$$\text{Var}[\varepsilon_i \mid x_i] = \mathbb{E}[\varepsilon_i^2 \mid x_i] = \sigma^2.$$

Combined with the **i.i.d.** sampling of $\{(x_i, \varepsilon_i)\}$, which leaves the errors conditionally uncorrelated across observations, this is the matrix statement

$$\text{Var}[\varepsilon \mid X] = \mathbb{E}[\varepsilon \varepsilon' \mid X] = \sigma^2 I_n.$$

$\Omega = \mathbb{E}[\varepsilon_i^2 x_i x_i']$, and if we assume **conditional homoscedasticity**:

$$\text{Var}[\varepsilon_i \mid x_i] = \mathbb{E}[\varepsilon_i^2 \mid x_i] = \sigma^2,$$

then by the law of iterated expectations

$$\Omega = \mathbb{E}[\varepsilon_i^2 x_i x_i'] = \mathbb{E}[\mathbb{E}[\varepsilon_i^2 \mid x_i] x_i x_i'] = \sigma^2 \mathbb{E}[x_i x_i'] = \sigma^2 Q.$$

The sandwich therefore collapses, $V = Q^{-1}(\sigma^2 Q)Q^{-1} = \sigma^2 Q^{-1}$, and the asymptotic normality becomes:

$$\sqrt{n}(\hat{\beta}^{\text{OLS}} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}),$$

which is the large-sample echo of the finite-sample **conditional variance** $\sigma^2(X'X)^{-1}$ derived below. Substituting $\hat{Q}^{-1} = n(X'X)^{-1}$ into the statistic gives

$$t_j = \frac{\sqrt{n}\hat{\beta}_j^{\text{OLS}}}{\sqrt{\hat{\sigma}^2 \cdot n[(X'X)^{-1}]_{jj}}} = \frac{\hat{\beta}_j^{\text{OLS}}}{\hat{\sigma} \sqrt{[(X'X)^{-1}]_{jj}}},$$

whose denominator is the usual **standard error** of $\hat{\beta}_j^{\text{OLS}}$.

(iii) Finite Sample Conditional Variance of OLS Estimator

Theorem

$$\text{Var}[\hat{\beta}^{\text{OLS}} | X] = (X'X)^{-1} X' \text{Var}[\varepsilon | X] X (X'X)^{-1},$$

and under **homoscedasticity** this collapses to

$$\text{Var}[\hat{\beta}^{\text{OLS}} | X] = \sigma^2 (X'X)^{-1}.$$

Proof

Conditional on X , the matrix $(X'X)^{-1}X'$ is a constant, so it passes through the variance operator on both sides:

$$\begin{aligned} \text{Var}[\hat{\beta}^{\text{OLS}} | X] &= \text{Var}[\beta + (X'X)^{-1}X'\varepsilon | X] \\ &= (X'X)^{-1}X' \text{Var}[\varepsilon | X] X (X'X)^{-1}. \end{aligned}$$

Substituting $\text{Var}[\varepsilon | X] = \sigma^2 I_n$,

$$\begin{aligned} \text{Var}[\hat{\beta}^{\text{OLS}} | X] &= \sigma^2 (X'X)^{-1} X' X (X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1}. \end{aligned}$$

(iv) Unbiased Estimator for the Error Variance

We still need $\hat{\sigma}^2$. For simplicity, we derive the consistent estimator for the variance, $\hat{\sigma}^2$, and then take the square root of it to obtain the consistent estimator for the standard deviation, which is guaranteed by the **continuous mapping theorem**.

A possible estimator $\hat{\sigma}^2$ is:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n e_i^2 = \frac{1}{n} e' e$$

where $e = y - X\hat{\beta}^{\text{OLS}}$.

However, the problem is $\mathbb{E}[\frac{1}{n}e'e | X] = \frac{n-k}{n}\sigma^2 \neq \sigma^2$, i.e., a biased estimator. Although we care more about the consistency, the following variance estimator is commonly used as it is not only consistent but also unbiased.

Theorem

$\frac{1}{n-k}e'e$ is an unbiased estimator for σ^2 .

Proof

We should first connect $\mathbb{E}[e'e]$ with ε , and y is closely related to ε , so we can start from here:

$$\begin{aligned}e &= y - X\hat{\beta}^{\text{OLS}} \\&= y - X(X'X)^{-1}X'y \\&= (I - P)y \\&= My \\&= M(X\beta + \varepsilon) \\&= M\varepsilon\end{aligned}$$

Note that M is the **residual maker** spanned from X , so $MX = 0$.

$$\begin{aligned}\mathbb{E}[e'e \mid X] &= \mathbb{E}[\varepsilon'M'M\varepsilon \mid X] \\&= \mathbb{E}[\varepsilon'M\varepsilon \mid X] \\&= \mathbb{E}[\text{tr}(\varepsilon'M\varepsilon) \mid X] \\&= \mathbb{E}[\text{tr}(M\varepsilon\varepsilon') \mid X] \\&= \text{tr}(M\mathbb{E}[\varepsilon\varepsilon' \mid X]) \\&= \text{tr}(M)\sigma^2\end{aligned}$$

The second equality is because M is **symmetric and idempotent** and the third one is due to the scalar value of $\varepsilon'M\varepsilon$ so $\varepsilon'M\varepsilon = \text{tr}(\varepsilon'M\varepsilon)$. The second-last equality is because both tr and $\mathbb{E}$ are linear operators, and the last one substitutes $\mathbb{E}[\varepsilon\varepsilon' \mid X] = \sigma^2 I_n$, which is exactly the **homoscedasticity** assumption.

$$\begin{aligned}\text{tr}(M) &= \text{tr}(I - P) \\&= n - \text{tr}(X(X'X)^{-1}X') \\&= n - \text{tr}((X'X)^{-1}X'X) \\&= n - k \\ \Rightarrow \mathbb{E}[e'e \mid X] &= \frac{n-k}{n-k}\text{tr}(M)\sigma^2 = (n-k)\sigma^2\end{aligned}$$

Therefore, the unbiased estimator of σ^2 is $\frac{1}{n-k}e'e$.

The divisor $n - k$ rather than n is the **degrees-of-freedom** count: the k **moment conditions** $X'e = 0$ are constraints the residuals satisfy for every dataset, leaving

$n - k$ free residuals.

(v) Consistent Estimator for the Error Variance

Theorem

Assuming *homoscedasticity*, $\frac{1}{n-k}e'e$ is a consistent estimator for σ^2 :

$$\frac{1}{n-k}e'e \xrightarrow{p} \sigma^2$$

Proof

Starting again from $e = M\varepsilon$ with $M = I - P$,

$$e'e = \varepsilon' M \varepsilon = \varepsilon' \varepsilon - \varepsilon' P \varepsilon,$$

so

$$\frac{1}{n-k}e'e = \frac{n}{n-k} \left[\frac{1}{n} \varepsilon' \varepsilon - \frac{1}{n} \varepsilon' P \varepsilon \right].$$

The factor $\frac{n}{n-k} \rightarrow 1$ because k is fixed.

For the first term, $\{\varepsilon_i^2\}$ are i.i.d. with $\mathbb{E}[\varepsilon_i^2] = \sigma^2$ under homoscedasticity, so by *Khinchine's WLLN*,

$$\frac{1}{n} \varepsilon' \varepsilon = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 \xrightarrow{p} \sigma^2.$$

For the second term,

$$\frac{1}{n} \varepsilon' P \varepsilon = \left(\frac{X' \varepsilon}{n} \right)' \left(\frac{X' X}{n} \right)^{-1} \left(\frac{X' \varepsilon}{n} \right),$$

where $\frac{X' \varepsilon}{n} \xrightarrow{p} 0$ and $\left(\frac{X' X}{n} \right)^{-1} \xrightarrow{p} Q^{-1}$ are both *already established*. Hence by the *continuous mapping theorem*,

$$\frac{1}{n} \varepsilon' P \varepsilon \xrightarrow{p} 0' Q^{-1} 0 = 0.$$

Therefore,

$$\frac{1}{n-k}e'e \xrightarrow{p} \sigma^2 - 0 = \sigma^2.$$

(vi) Test Statistic

Finally, putting the **consistent variance estimator** into the statistic derived above, the test statistic is

$$t_j = \frac{\hat{\beta}_j^{\text{OLS}}}{\sqrt{\frac{1}{n-k} e' e \cdot \sqrt{[(X'X)^{-1}]_{jj}}}} \xrightarrow{d} N(0, 1)$$

under $H_{0,j} : \beta_j = 0$.